



The Web of Worlds

W3C Immersive Web Group, F2F Meeting, NYC, May 19th + 20th

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Chairs: Metaverse Standards Forum ([Metaverse-standards.org](https://metaverse-standards.org))

Members: Web3D Consortium ([Web3D.org](https://web3d.org)): Official Liaison: W3C, ISO, OGC, Khronos)



Metaverse

STANDARDS FORUM™

**Standards Cooperation for an Open
and Inclusive Metaverse**

metaverse-standards.org | [@Metaverse_Forum](https://twitter.com/Metaverse_Forum)

Motivation and Problem Statement

- The Metaverse combines the **connectivity of the Web** with the immersiveness of Spatial Computing.

*How can we **connect** spatial data, user, devices and apps?*

- Use cases demonstrate a wide variety of ways **that virtual worlds** and physical worlds may be linked.

How can we distinguish these different modes and their requirements?





Opportunity

Convergence:

- Ripple effects from Web3D Consortium + W3C Declarative 3D Community Group with new fuel from the Metaverse Standards Forum
- New technology realities and opportunities
- Leverages a proven open standard and open source ecosystem:
 - W3C: HTML5, CSS, URI, WASM, WebGPU, WebXR, ...
 - Web3D: X3D, HANIM
 - Khronos: WebGL, Vulkan, gltf/glb, OpenXR, ...
 - OGC: Geopose, 3DTiles, ...
 - IANA: model/* types



New Result

<https://webofworlds.github.io/>

- The **3D Web Interoperability Group** in the Metaverse Standards Forum has worked through the Usability Engineering process, developing the open-source, royalty-free, **Web of Worlds** Whitepaper and API proposal:
 - Use Cases
 - Requirements
 - Gap Analysis
 - Whitepaper, github, and website
 - ***Blog forthcoming*** as work moves to the Spatial Computing WG
- The result is a clean API service architecture and service stack that leverages existing W3C, Web3D, Khronos, IANA, and OGC standards.



The Web of Worlds (WoW) Architecture and API

To reduce complexity and encourage adoption, WoW leverages a harmonized "glue layer" of existing standards. The architecture is built on three core pillars:

1. **Open Spatial World API**: Treats virtual worlds as persistent **API and HTTP endpoints**, allowing users to join, view, and preview worlds using a single URI.
2. **Open User Manifest API (The "Digital YOU")**: Powers a **portable, user-controlled identity** layer. Utilizing **JSON-LD, Decentralized Identifiers (DIDs)**, and Verifiable Credentials, users can carry their avatars, preferences, and assets across different virtual worlds.
3. **Open Spatial Asset API**: Facilitates modular, reusable 3D components by leveraging existing standards such as **glTF, X3D, USD**, and others, which are already registered with IANA.



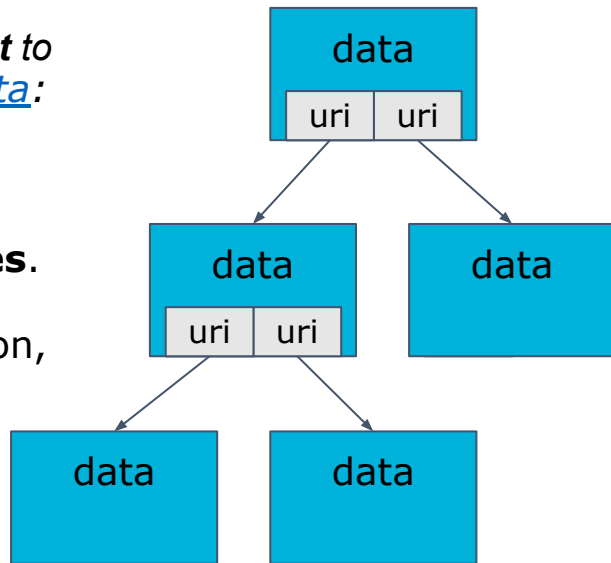


Scope: ***The Connectivity of the Web***

Connectivity of the Web

Defining Transformation in last 30 years: from **web of hypertext** to the **web of data**, [Tim Tim Berners-Lee, 2006-07-27, Linked Data:](#)

1. Use **URIs** as **names** for **things**
2. Use **HTTP URIs** so that people can **look up** those **names**.
3. When someone looks up a **URI**, provide useful information, using **standards**
4. Include links to other **URIs**. so that they can **discover** more things.



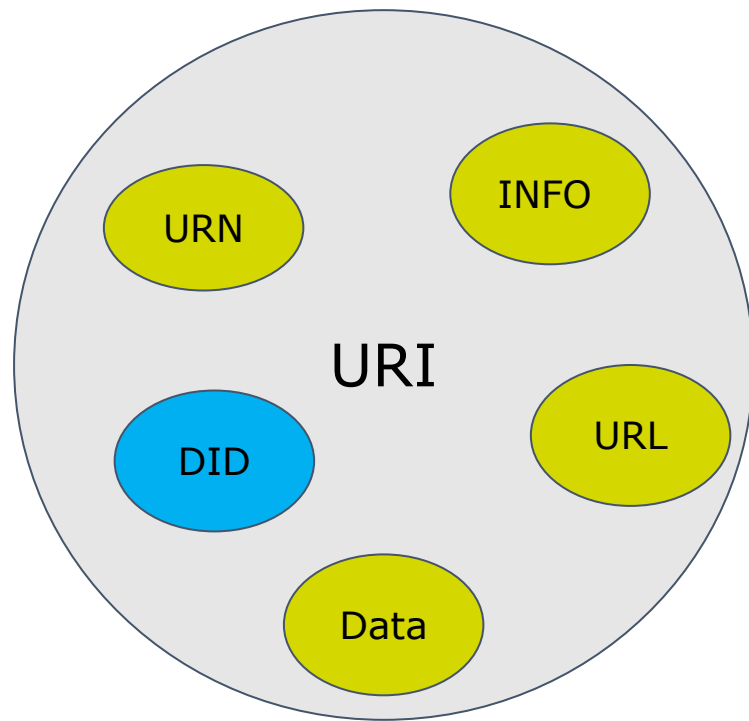
=> The modern web is more relaxed in how data semantics and data structure are related to individual data instances.

For example, there are external standards (e.g., ISO for X3D and JT), JSON-LD context, and RDF as existing patterns that all participate in this same network.

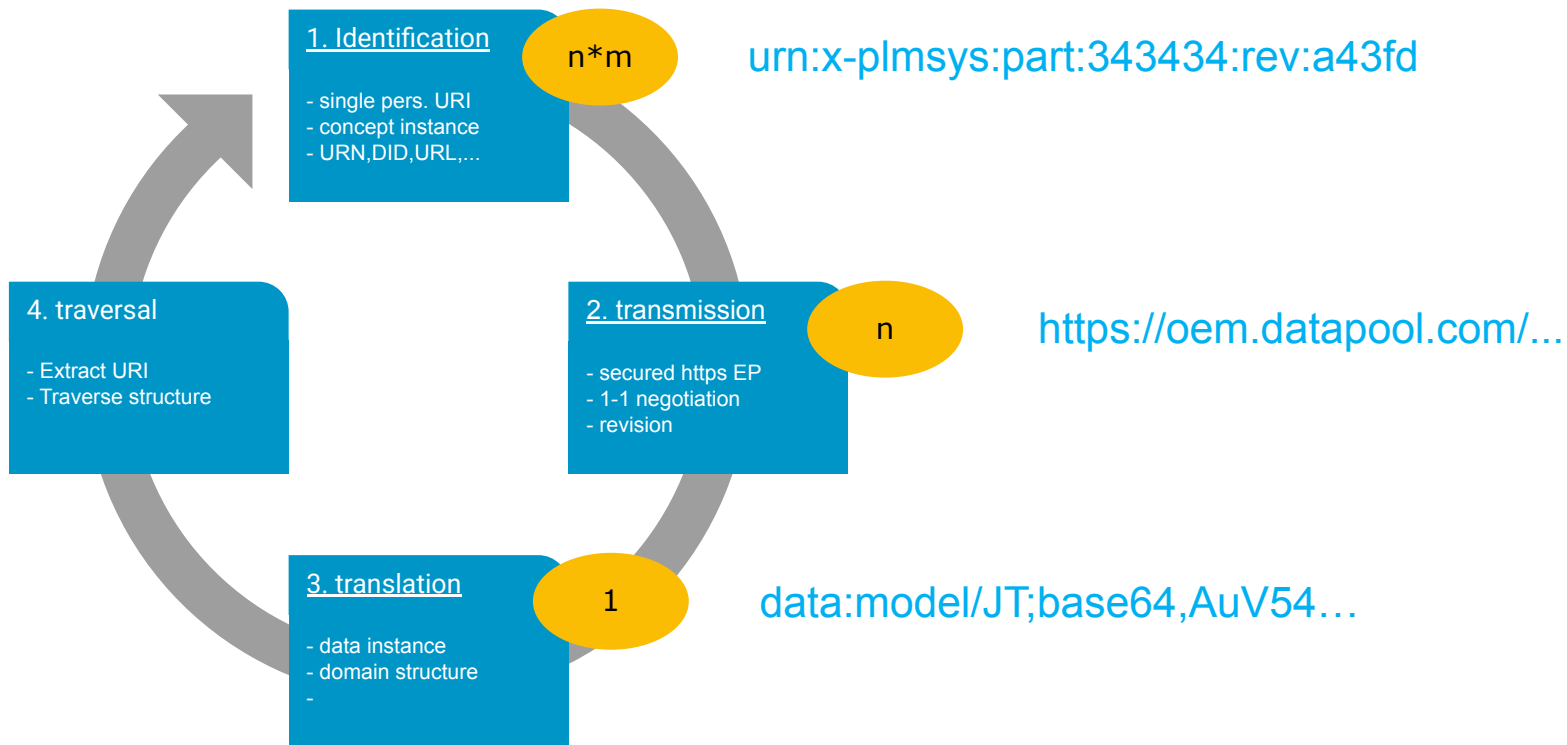
Universal Resource Identifier

URIs are abstract concepts which have a growing number of derived standards as implemented:

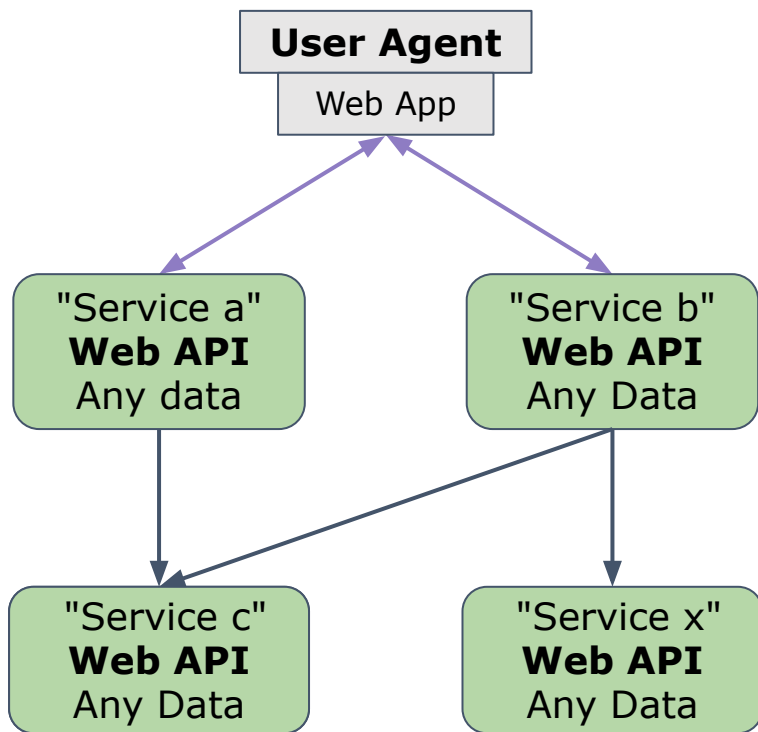
- **URN**: "name" through namespaces
 - urn:uuid:..., urn:urn:x3d:...; urn:x-i3d:...
- **INFO**: "information assets" through namespaces
 - info:oclnum:...; info:dewey:...
- **DID**: "identify" through methods
 - did:sov:..., did:bctr..., did:key:
- **URL**: "locate" through schemes
 - http://asfddata:model/JT;base64,AuV54...
- **DATA**: "
 - data:model/JT;base64,AuV54...



Multi-stage and decentralized web data

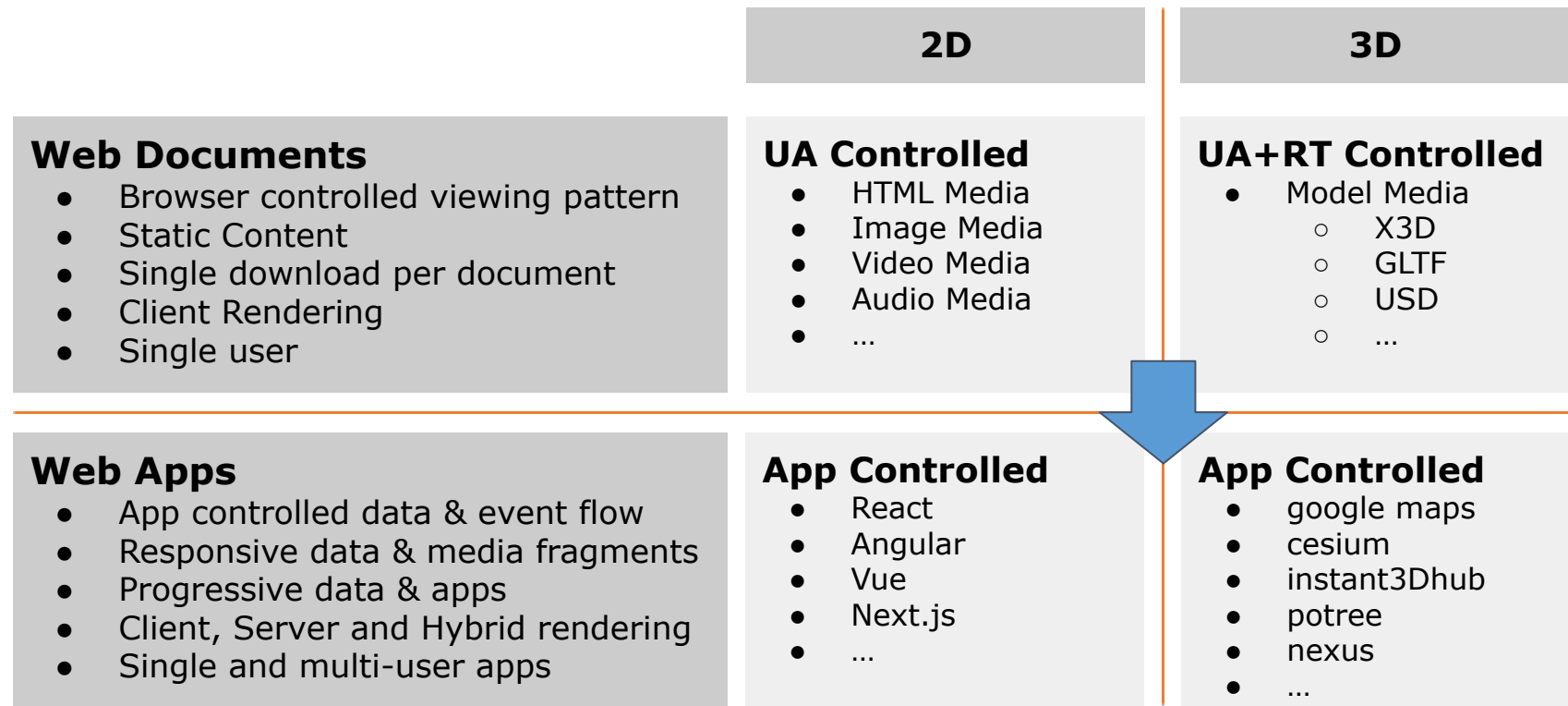


The (modern) Web

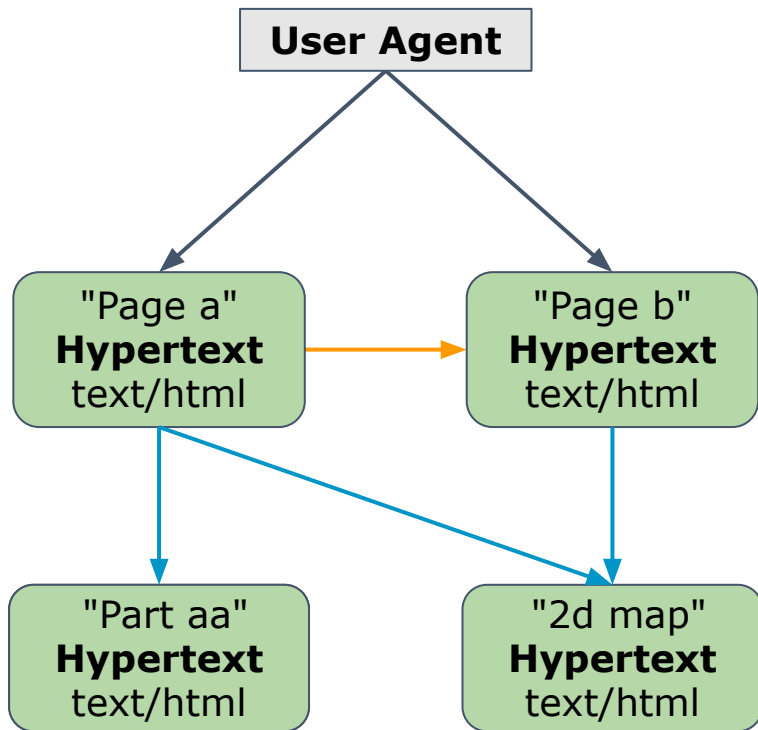


1. **From Web Documents to Web Apps**
 - a. App controlled data & event flow
 - b. Responsive data & media fragments
 - c. Progressive data & apps
 - d. Client, Server and Hybrid rendering
 - e. Single and multi-user apps
2. **From Web Resources to API Endpoints**
 - a. HTTP & WS data transmission
 - b. 1To1 content negotiation and encoding
 - i. html/text
 - ii. application/json
 - iii. model/jt
 - iv. ...
 - c. Service agnostic authentication (e.g. SSO)
3. **From Last mile to Service Networks**
 - a. Deep service dependency stack (> 30)
 - b. Microservice lead to smaller packages
 - c. Native- and Web-Client symmetry

The modern 3D Web continuum

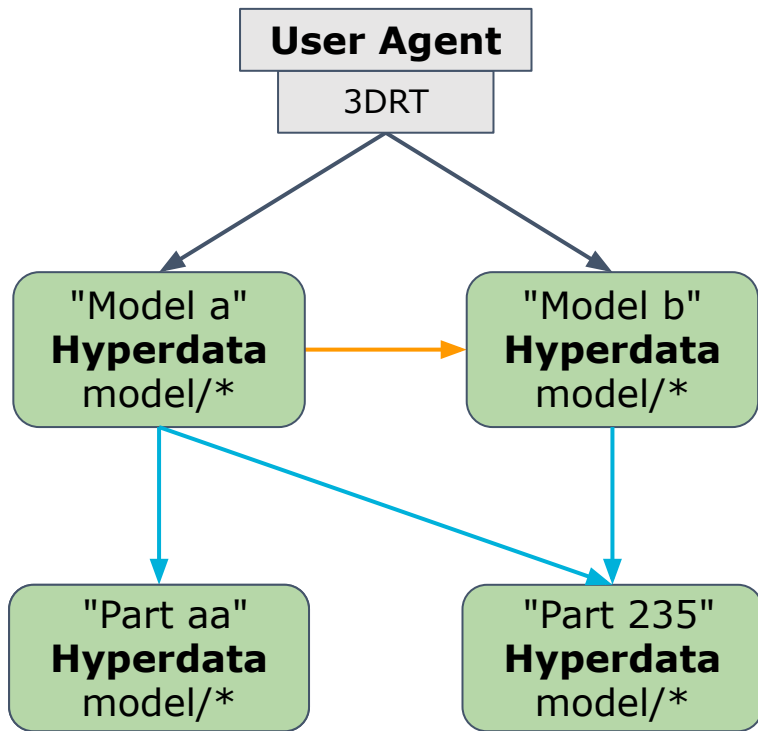


Connectivity of the (classic) 2D Web



- **URL** are pointer to **HTTP Resources** which can be of any **content** type ([http 1.0](http://1.0))
- **text/html** as **Hypertext** content
 - Is evaluated in the **User-Agent**
 - In domain URI references
 - `<a>` Links to new html-page
 - `target='_blank'`
 - `<iframe>` Links to sub-page

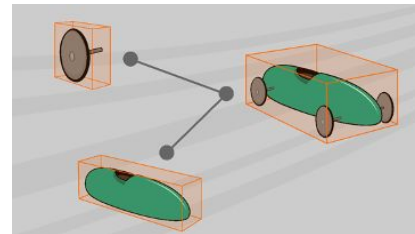
Connectivity of the (classic) 3D Web



- **URL** are pointer to **HTTP Resources** which can be of any **content** type ([http 1.0](http://1.0))
- **Any model/*** as **3D content** data
 - Is evaluated in a **3D Runtime (3DRT)**
 - **Browser Plugin**
 - **Javascript Frameworks**
 - **Three.js**
 - **X3DOM**
 - **X_ITE**
 - ...
- **RFC2077** defines the model domain
 - **spatial coordinate system**
 - **In domain references**
 - **<...>** Links to new scene
 - **<...>** Links to sub-data
- >40 **IANA-Registered** types
 - **model/gltf-binary**
 - **model/vnd.usda**
 - **model/x3d+xml**
 - ...

Connectivity of the (classic) 3D Web

- Links inside the (model) domain
 - Builds on existing HTTP standards
 - Implemented in existing standards



In Domain Links	None	In-Format	Any-Format
Format	GLTF, obj, stl, ...	USD, GLTFX, collada, plmxml, ...	X3D, STEP, ...

<X3D>

<Transform>

<Inline url='http://example.com/spatialpool.cgi?id=3478902'/>

</Transform>

</X3D>

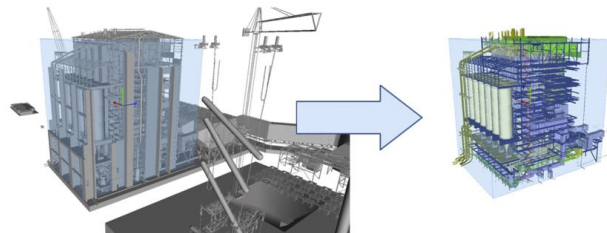
(Valid X3D example pointing to data endpoint with undefined format.
The model format will be negotiated in the actual API call)

Connectivity of the (classic) 3D Web

- **Links inside the (model) domain and fragments**

- URI Fragment standards allow to address sub-data

E.g.: [RFC7111](#) for "text/csv":
"example.com/data.csv#row=3-5"



W3C Media Fragment URI

Spatially-, temporally- and structure-based addressing schema

example.com/media/movie.mpg#xywh=100,100,10,10&t=10,20

example.com/media/movie.mpg#id=cap

Open Opportunity: Missing "Model Fragment URI" ***Spatially-, temporally- and structure-based addressing***

E.g., example.com/model/434/#xyzwhz=100,100,100,20,20,20&t=10,20

Object- and Object-set-based addressing schema

example.com/model/434/#objects="/tire"



Vision: The Web of Worlds

Mechanisms to link and reference virtual worlds and parts thereof

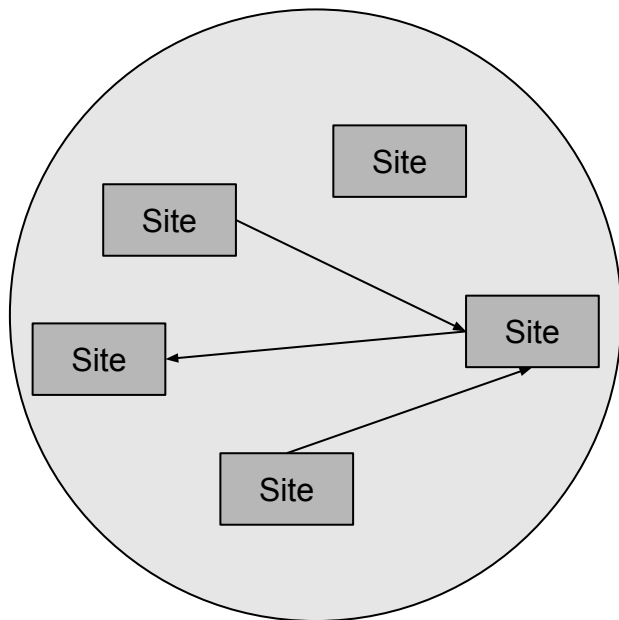
[WebOfWorlds | home page](#)

<https://webofworlds.github.io/>

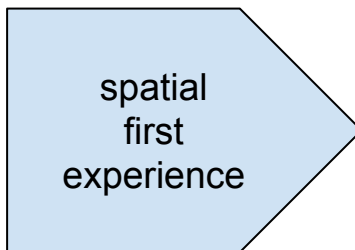
Vision: The web of worlds

Web

Site is HTTP endpoint

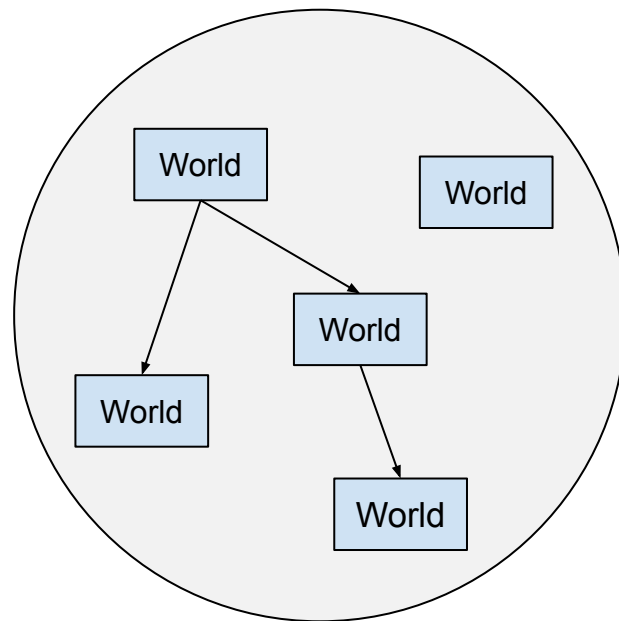


2D > 2D > 2D > 2D



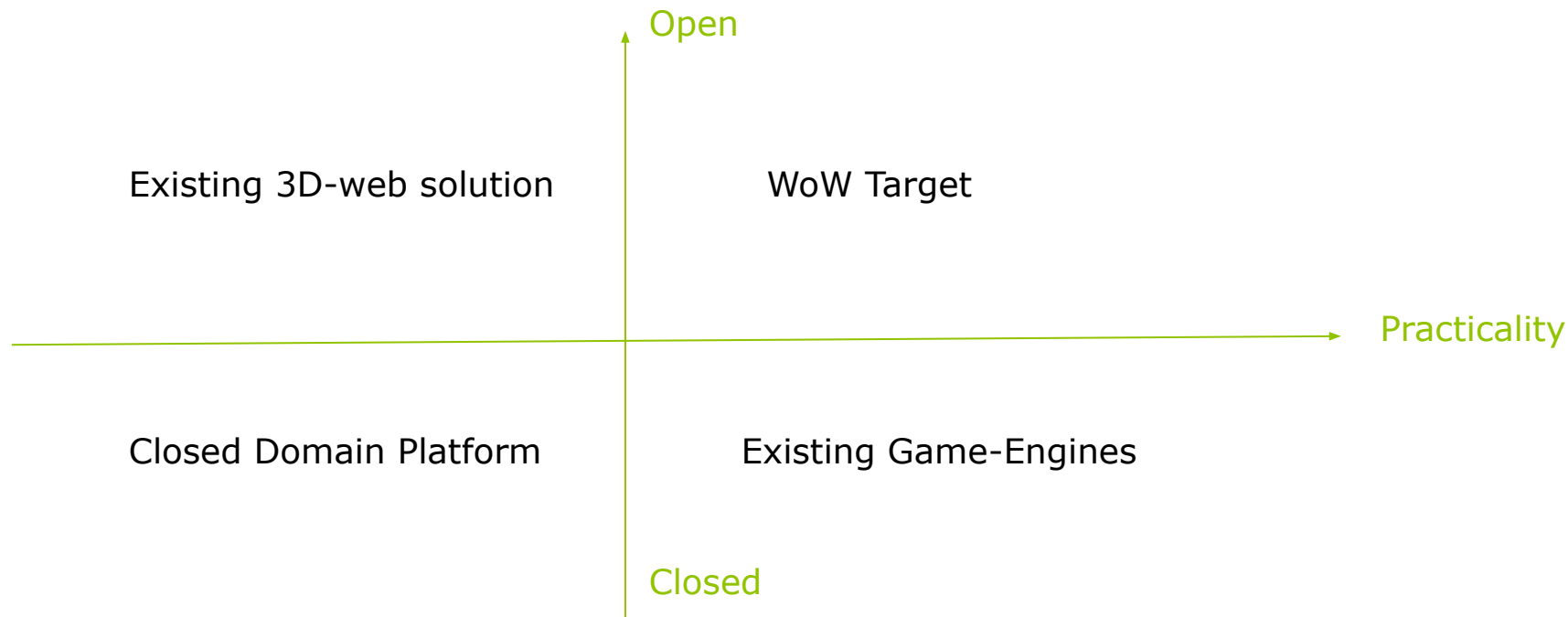
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World is API & HTTP endpoint



3D > 3D > 3D > 3D

Issue: Exposed complexity kills practicality



Important shift: from shared apps to shared worlds

1. Single URI pointing to discrete virtual world

- a. **Links between worlds** (e.g jumps/portals and inserts in spatial composition)
- b. **Persistent data references in linked data environments** (e.g. digital twins, product configuration, ...)
- c. **Storing** URI/URL for later use (e.g. Bookmarks, Ticket system, ...)
- d. **Sharing** URI with second user (e.g. teams, ...)



2. Can be opened and joined in any browser directly

- a. Shared interactive **spatial experience** and not 3D data (e.g x3d, gltf, usd)
- b. **Web app** controlled **user experience** as dominant delivery model (e.g client data vs remote rendering)

3. Rich user experience for any spatial data composition and size

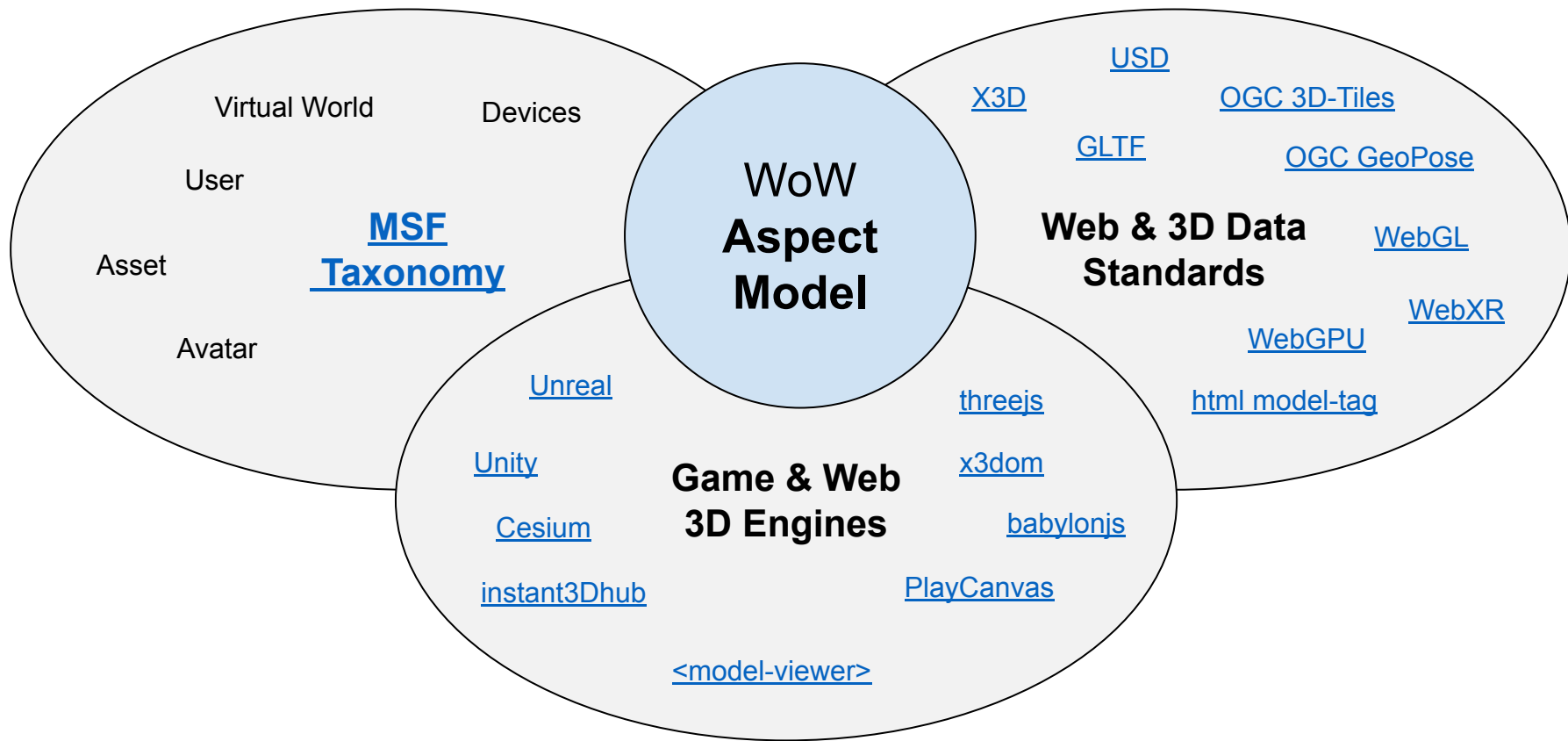
- a. **Static and dynamic spatial data composition**, billions of addressable spatial data states
- b. **High visual efficiency and fidelity**

4. Shared multi-user and multi device scenarios

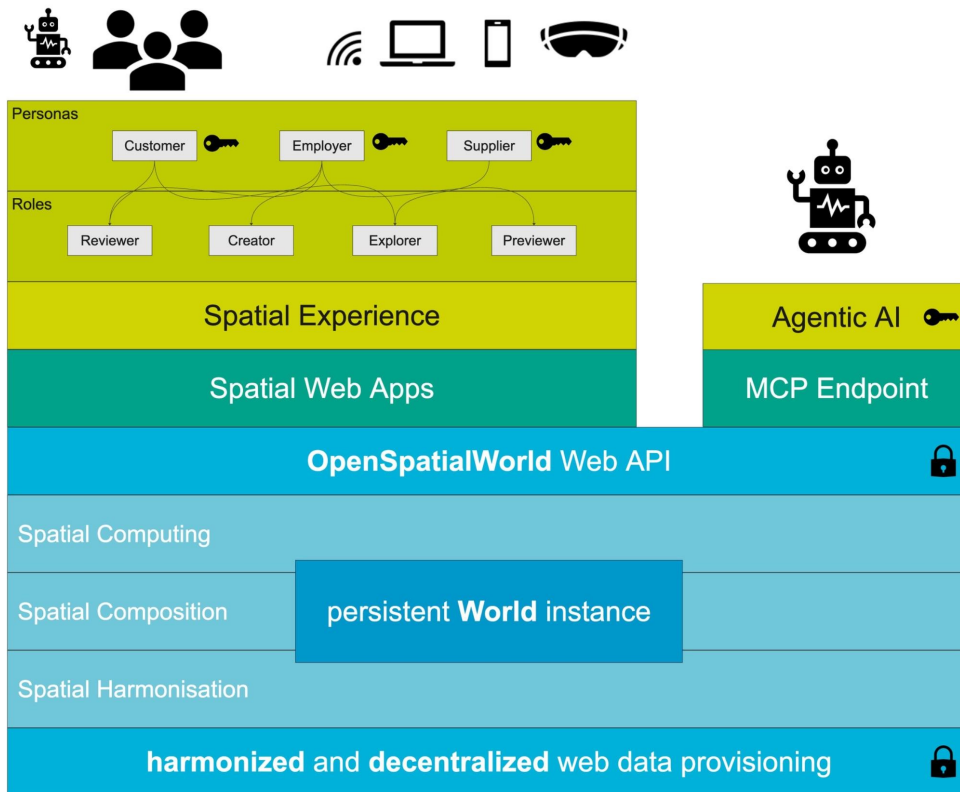
- a. Mixed and highly dynamic user and device configurations (e.g desktop, mobile, immersive)

5. World-agnostic user identification and data authentication (e.g. SSO)

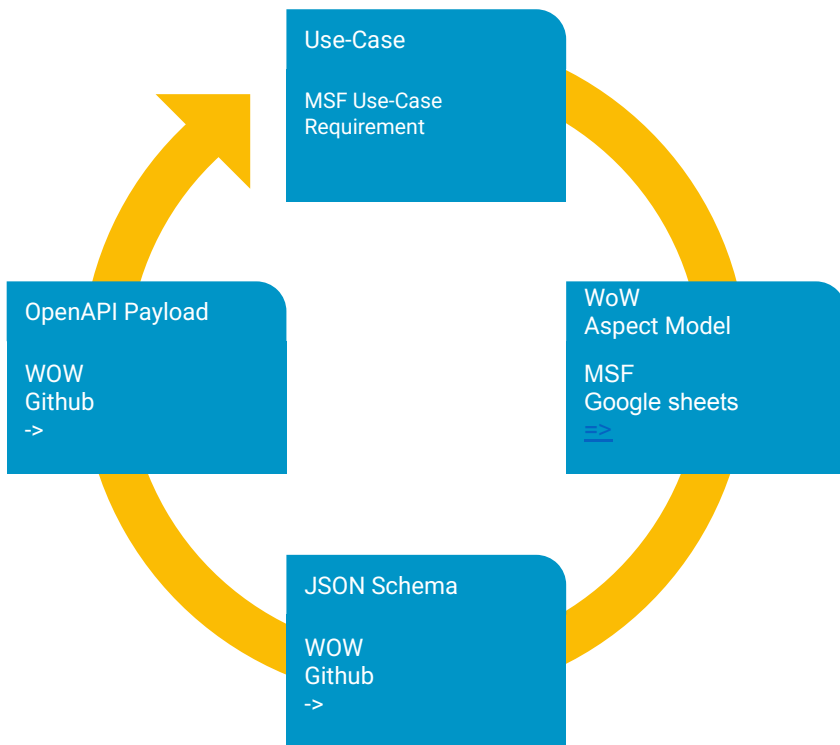
Important shift: Unified and simple "WoW Aspect Model"



WoW WEP API for Apps and Agents



Continuous WoW Aspect Model validation



World	Name	type	description
	label	string	
	age_restriction	number	
	license	string	
	cost	string	
	revision	number	
	duration	seconds	
Presence	Name		
	avatar	URI	
	navigation	string	valid navigation mode (e.g. fly)
	gravity	number	
Tehnology	Name		
	Device-class	enum	desktop, mobile webxr--immersive
	Runtime	string	e.g. UNREAL_WebGPU_51, x3domWorld,
Users	Name		
	active_user_count	number	Number of active
	total_user_count	number	Total number of user with unique ID
Viewa	Name		
	view_count	number	Number of available views with unique ID
Portals	Name		
	view_count	number	Number of available portal with unique ID



Github: WoW API and Aspect Schema

- <https://github.com/WebOfWorlds/WOWAPI>

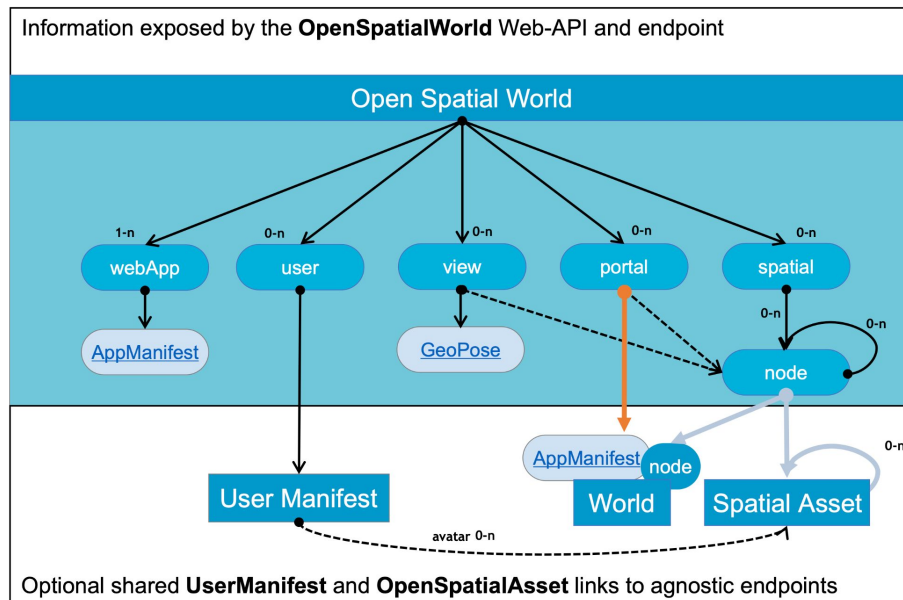
- **WOW-API**

- **OpenSpatialWorld-API.**

- OpenAPI spec
 - World Aspects
 - webApp JSON Schema
 - User JSON Schema
 - View JSON Schema
 - Portal JSON Schema
 - Node JSON Schema

- **OpenSpatialAsset-API**

- OpenAPI spec
 - Asset Aspects
 - Asset JSON Schema



CORE API: scene node CRUD cycle

	<u>Create</u> +Assets Links	<u>Create</u> +Asset Copy	<u>read</u>	<u>update</u> +Assets Links	<u>update</u> +asset copy	<u>delete</u>
REST WOW	/node/ POST => JSON-Tree <= JSON-Tree with new nodeID	/node/ POST => model/* asset	/node/ GET =>fields-white list + depth <= JSON-Tree	/node/ Update => JSON-Tree + depth	/node/ Update	/node/ Delete => NodeID
MCP	Tools createNodes => JSON-Tree	? binary data in Tools	resource	Tool updateNodes => Json-Tree	? binary Data in tools	Tool deleteNodes => NodeID group

JSON Node Payload

```
[
  {
    "id": 0,
    "name": "string",
    "tags": [
      "string"
    ],
    "parent": 0,
    "children": [ { } ]
    "localTransform": [
      0
    ],
    "spatialAssetURI": "string",
    "appearanceURI": "string"
  }
]
```

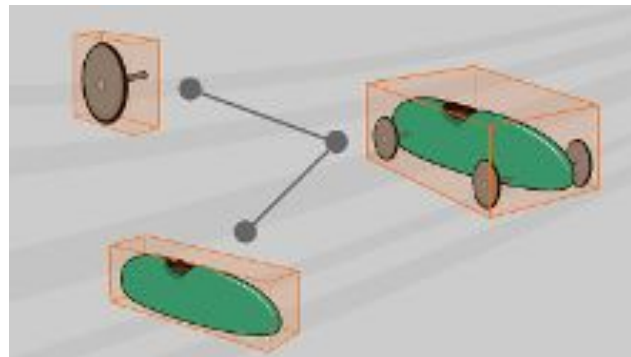


Example: create nodes with linked assets

<http://example.com/world/604djq942/wow/scene/node/35643>

POST

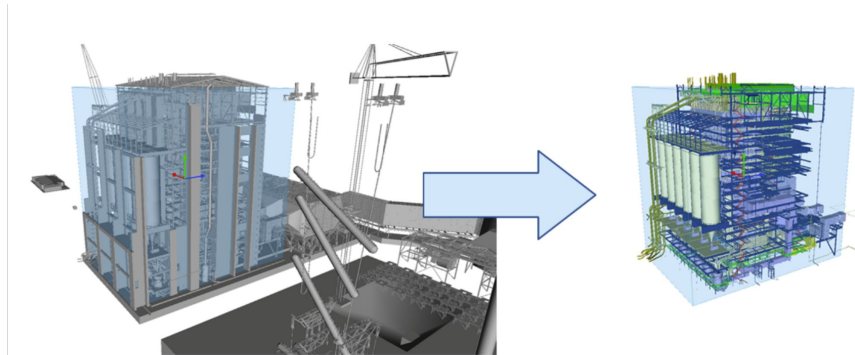
```
{
  "name": "body",
  "assetURI": "http://example.com/data/carbody056834",
  "children": [
    {
      "name": "leftFront"
      "localTransform": [ ... ]
      "assetURI": "did:example:123?service=wheelAssembly35"
    },
    {
      "name": "rightFront"
      "localTransform": [ ... ]
      "assetURI": "did:example:123?service=wheelAssembly35"
    },
    {
      "name": "leftBack"
      "localTransform": [ ... ]
      "assetURI": "did:example:123?service=wheelAssembly35"
    },
    {
      "name": "rightBack"
      "localTransform": [ ... ]
      "assetURI": "did:example:123?service=wheelAssembly35"
    }
  ]
}
```



2 Assets/URI with 5 data instances

Spatial computation function on Worlds

- **Compute Bounding Volume**
 - < bounds
- **Compute Volume Selection**
 - < bounding volume
 - > node selection
- **Compute Minimal Distance**
 - < nodeGroupA, nodeGroupB
 - > minimal Distance
- **Compute Clashes / Collision**
 - < nodeGroupA, nodeGroupB
 - > clasList
- **Compute Preview**
 - < user, view or portal
 - > model/image binary



Spatial Compute API: Restful and/or MCP Endpoint ?

	Compute Bounding Volume	Compute Volume Selection	Compute Minimal Distance	Compute Clashes	Compute Preview	... s
REST WOW	??	??	??	??	??	??
MCP	Tool spec	Tool spec	Tool spec	Tool spec	Tools spec	Tool spec

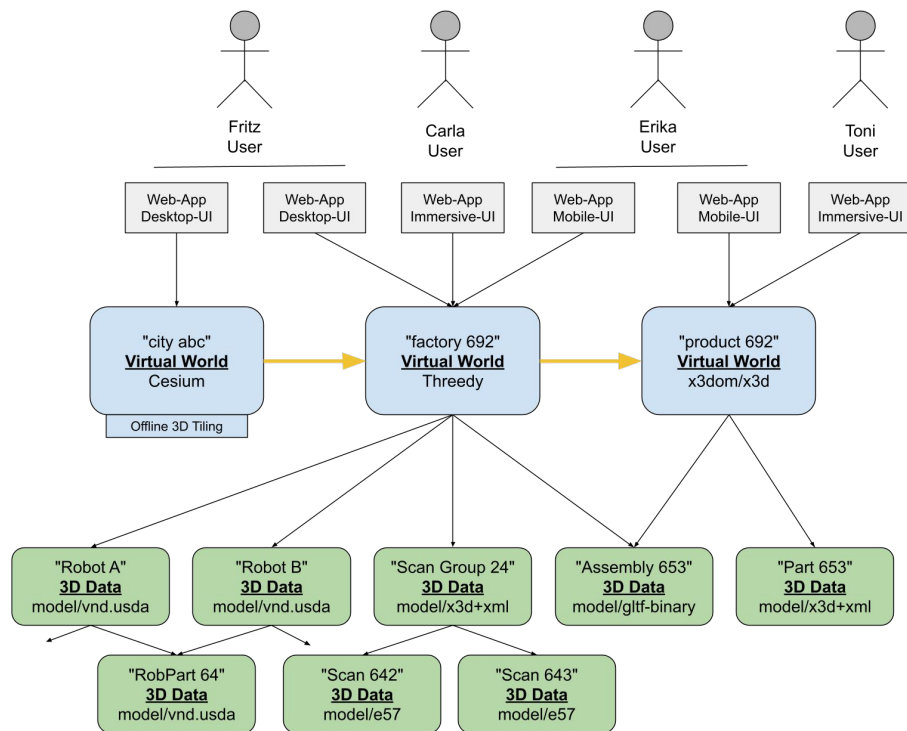


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Examples

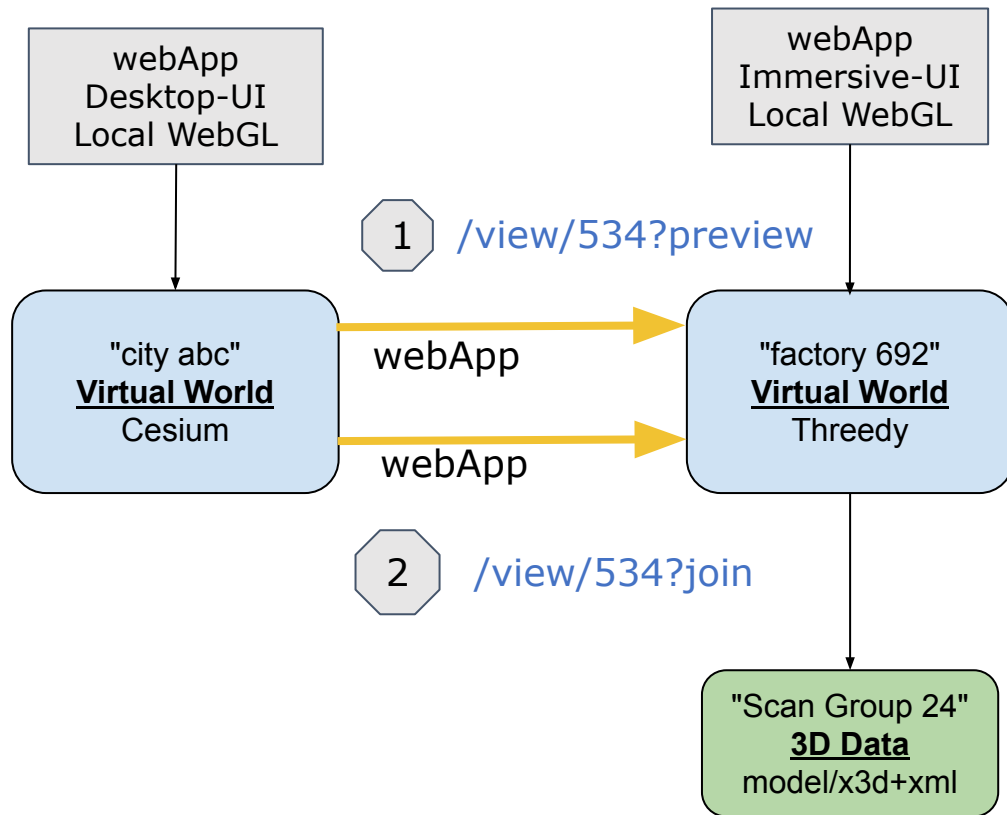


Example 1/3: Shared links in multi-user/device setups

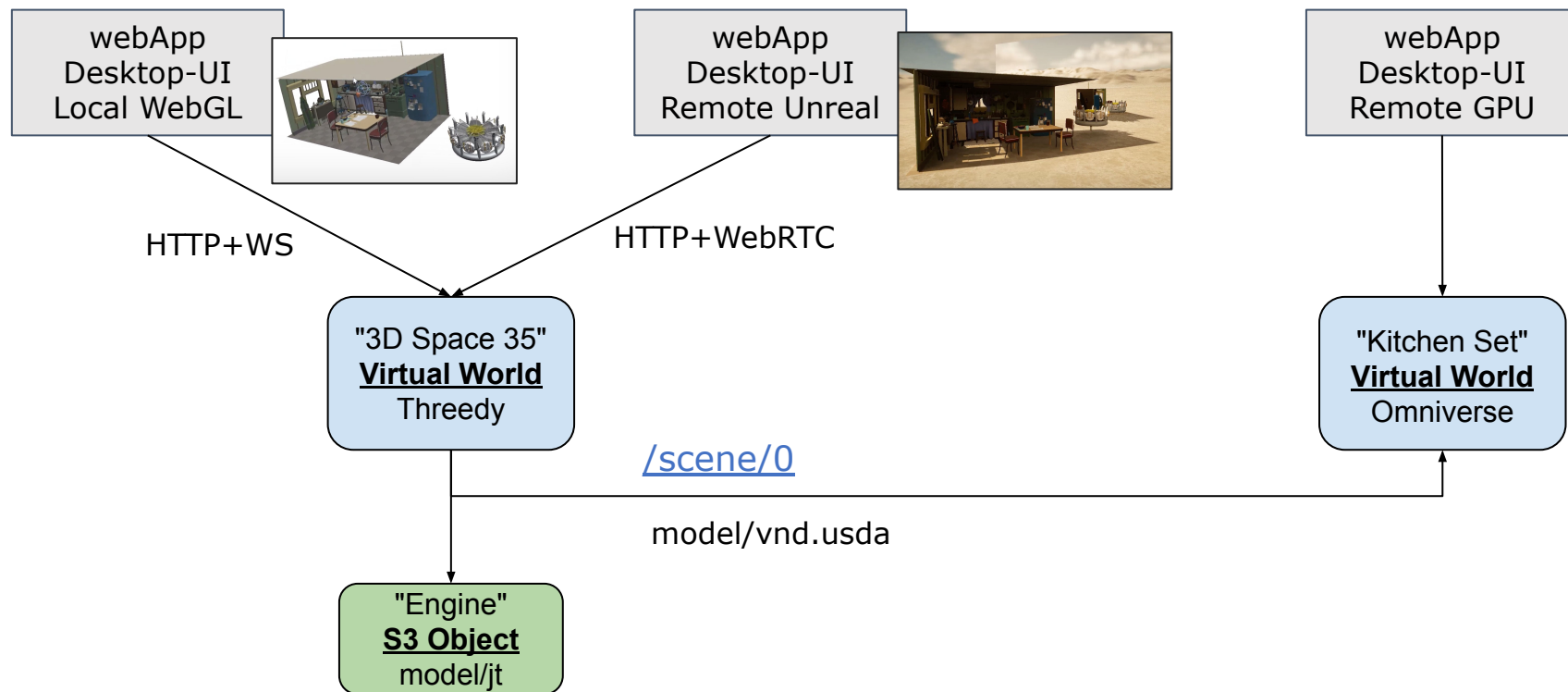


- **Fritz** clicks "city_abc" link in email
 - UA starts new Web-App with desktop-UI
- **Fritz** navigates to factory section
- **Fritz** clicks on "explore" button
 - UA starts new Web-App with desktop-UI
- **Fritz** shares "factory 692" link with Carla
- **Carla** Opens "factory 692" link on XR-Class
 - UA starts new Web-App with immersive-UI
- **Carla** shares link with Erika in teams
- **Erika** Open "factory 692" link on phone
 - UA starts new Web-App with mobile-UI
- **Erika** navigates to product section
- **Erika** activates "product" objects
 - UA starts new Web-App with mobile-UI
- **Erika** shares the "product 692" link with Toni
- **Toni** open "Product 692" on AR-class
 - UA starts new Web-App with immersive-UI

Example 2/3: Previews and portals



Example 3/3: Compose scene state from second world





Roadmap to Interoperability

webofworlds.org

Test implementations: @ SIGGRAPH 2026 and Web3D 2026

Advocate compatibility points across:

- W3C Standards
- Web3D Standards
- Khronos Standards
- OGC Standards
- ...

Next Steps

- New Home
 - webofworlds.org
- GitHub project and OpenAPI
 - <https://github.com/WebOfWorlds/WoWAPI>
- First Implementation previews in 2026-Q2
 - simpleWorlds (open-source)
 - threedy.io (commercial)



Slide backup

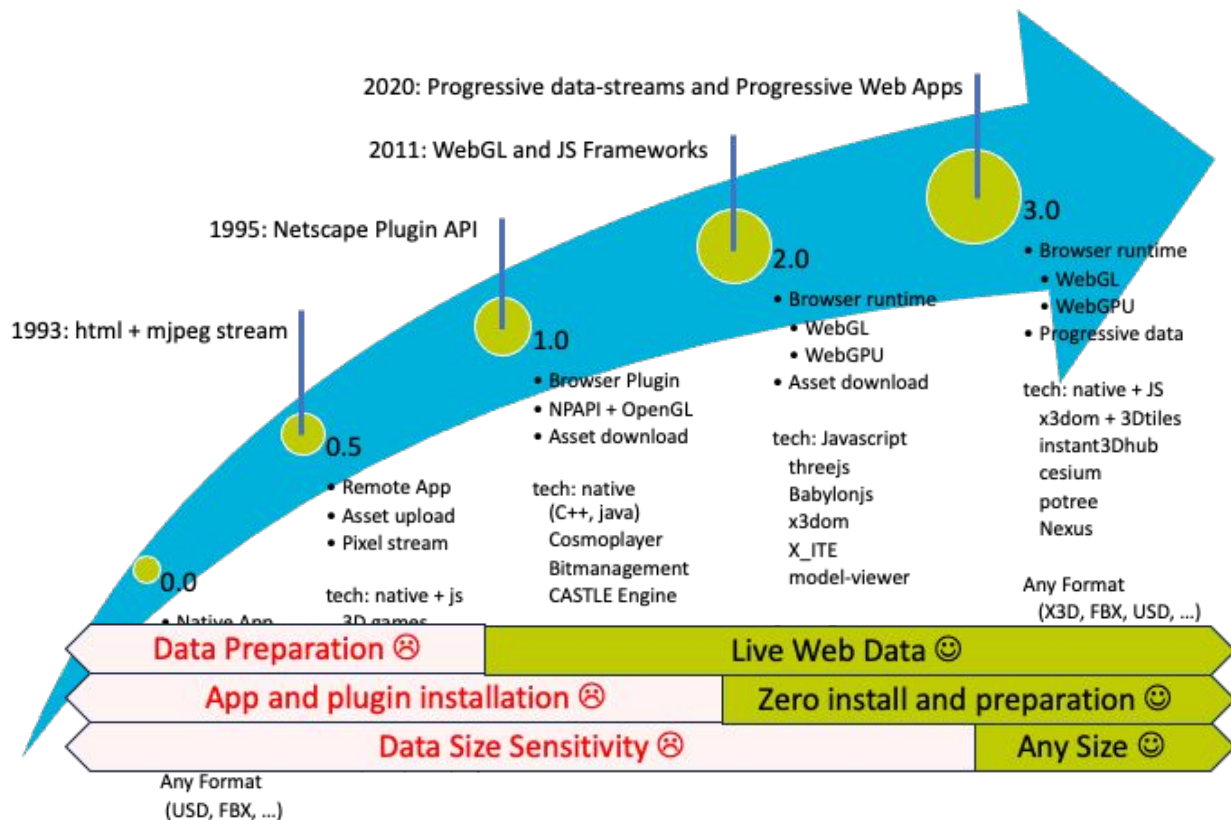
Validation: The Seven Rules of the Metaverse check

1. There is only one Metaverse. 
2. The Metaverse is for everyone. 
3. Nobody controls the Metaverse. 
4. The Metaverse is open. 
5. The Metaverse is hardware-independent. 
6. The Metaverse is a Network. 
7. The Metaverse is the Internet. 

■ ...



phases and evolution of 3D web



Existing Today: Rich technology and framework stacks

	2D Web App	3D Web App
Rich frameworks	React, Angular, Vue, Next.js	Three.js, x3dom, ...
Navigation	2 Page	6D Camera, WebXR Input
Multi User	Live sync UA	Live sync UA
Persistency	Online 2D Document	Online 3D Documents
Device Classes	Desktop, Mobile, (XR)	Desktop, Mobile, XR
Decentralized Identifiers	DID	DID
Security and Access delegation	SSO, OAuth	SSO, OAuth
Streaming	WebSockets, HTTP-Push	WebSockets, HTTP-Push
Local GPU Access	WebGL, WebGPU	WebGL, WebGPU

Why is the open web not the preferred platform today ?

